Reliable CMOS PUF Cell for IoT Hardware Security

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***Abstract*—This section will be completed after the simulation results and final comparison are added.**

***Index Terms*—Physically Unclonable Function (PUF), SRAM PUF, CMOS, hardware security, IoT, Schmitt trigger.**

I. Introduction

The rapid growth of Internet of Things (IoT) devices has significantly increased the demand for secure and reliable hardware authentication methods. Traditional security approaches rely on storing secret keys in non-volatile memory (NVM), making them vulnerable to physical attacks, cloning, and data extraction. As a result, Physically Unclonable Functions (PUFs) have emerged as an attractive hardware-based security solution.

A PUF generates a unique response by exploiting the inherent manufacturing variations of CMOS transistors. Since these process variations are random and unavoidable, each integrated circuit exhibits a unique physical characteristic that can be used as a hardware fingerprint. Unlike conventional memory-based security techniques, PUFs do not require permanent storage of secret keys, making them more resistant to cloning and tampering.

Among various PUF architectures, SRAM-based PUFs have received considerable attention because they utilize the natural power-up behavior of standard SRAM cells without requiring additional entropy sources. During power-up, each SRAM cell settles to either logic '0' or logic '1' depending on the slight transistor mismatches inside the cell. However, environmental conditions such as temperature, supply voltage variations, and electrical noise may cause some SRAM cells to generate unstable responses, reducing the reliability of the generated PUF.

In this project, a CMOS SRAM-based PUF is designed and implemented using Electric VLSI. The proposed architecture is based on a conventional 6T SRAM PUF enhanced with a Schmitt trigger-assisted readout circuit to improve the stability of the generated response. In addition, a 4×4 PUF array, an output latch, and an unstable-bit detection circuit are incorporated to evaluate and improve the reliability of the PUF output. The complete design is verified through schematic and layout implementation, followed by circuit simulation.

II. Related Work

SRAM-based Physically Unclonable Functions (SRAM PUFs) have been widely used for hardware authentication because they generate unique responses from the natural process variations of CMOS transistors. However, conventional SRAM PUFs may produce unstable responses under voltage, temperature, and noise variations.

Several techniques have been proposed to improve the reliability of SRAM PUFs. Among them, Schmitt trigger-assisted designs enhance the read stability by increasing the noise immunity of the sensing circuit, resulting in more reliable PUF responses.

Inspired by these approaches, this project implements a Schmitt trigger-assisted SRAM PUF using Electric VLSI. The proposed design includes a conventional 6T SRAM cell, a Schmitt trigger-assisted readout circuit, a 4×4 PUF array, an output latch, and an unstable-bit detection circuit to evaluate the reliability of the generated PUF response.

III. Proposed Design

A. Overall Architecture

The proposed system is designed to generate a reliable hardware fingerprint by exploiting the inherent process variations of CMOS transistors. The architecture consists of several integrated building blocks, including a CMOS inverter, a conventional 6T SRAM PUF cell, a Schmitt trigger-assisted readout circuit, a one-bit PUF readout unit, a 4×4 PUF array, an output latch, and an unstable-bit detection circuit. The CMOS inverter serves as the basic building block of the SRAM cell, while the SRAM cell acts as the entropy source that generates the PUF response during power-up. To improve the stability of the generated response, a Schmitt trigger-assisted readout circuit is employed before collecting the generated bits into the PUF array. Finally, the output is stored using an output latch and evaluated by the unstable-bit detection circuit to improve the reliability of the generated hardware fingerprint.

B. CMOS Inverter

The CMOS inverter is the fundamental building block of the proposed design and forms the basis for constructing the SRAM PUF cell. It consists of one PMOS transistor and one NMOS transistor connected in a complementary configuration. The inverter produces a logic-high output when the input is low and a logic-low output when the input is high, providing low static power consumption and high noise immunity. In this project, the inverter was first designed and

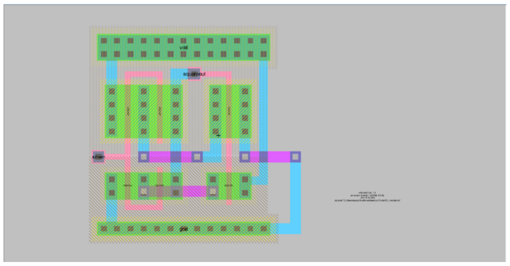

Fig. 1. Layout implementation of the proposed Schmitt trigger-assisted readout circuit.

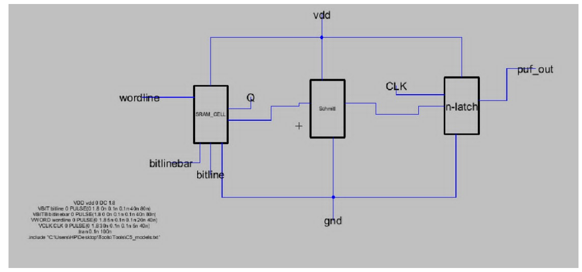

Fig. 2. Schematic implementation of the one-bit SRAM PUF readout circuit.

verified through schematic implementation, layout design, and circuit simulation before being integrated into the SRAM cell.

C. Conventional 6T SRAM Cell

The conventional 6T SRAM cell serves as the primary entropy source of the proposed PUF architecture. It is composed of two cross-coupled CMOS inverters and two access transistors controlled by the word line (WL). During power-up, the cell naturally settles to either logic '0' or logic '1' due to unavoidable transistor mismatches introduced during the fabrication process. These random but repeatable power-up states are utilized to generate the unique PUF response. The complete SRAM cell was implemented at both schematic and layout levels and verified through transistor-level simulation.

D. Schmitt Trigger-Assisted Readout Circuit

To improve the stability of the generated PUF response, a Schmitt trigger-assisted readout circuit is incorporated into the proposed design. The Schmitt trigger increases the noise immunity of the read operation by introducing different switching thresholds for rising and falling transitions. As a result, unstable voltage fluctuations around the switching point are significantly reduced, leading to a more reliable PUF response during repeated evaluations.

Figure 1 shows the layout implementation of the Schmitt trigger-assisted readout circuit designed using the Electric VLSI Design System. The layout was completed following the corresponding schematic and verified through DRC and LVS before being integrated into the proposed SRAM PUF architecture.

E. One-Bit PUF Readout Circuit

The one-bit readout circuit is responsible for extracting the response generated by an individual SRAM PUF cell. It provides the interface between the SRAM cell and the remaining blocks of the system, ensuring that the generated response can be accurately observed during simulation. This circuit serves as the basic readout element before extending the architecture to the complete PUF array.

Figure 2 illustrates the one-bit PUF readout circuit. The circuit combines the conventional SRAM cell with the Schmitt trigger readout stage to extract a stable PUF response. During the read operation, the stored value generated by the SRAM cell is processed through the Schmitt trigger before being forwarded to the output, providing a reliable single-bit response that serves as the basic building block for the complete 4×4 PUF array.

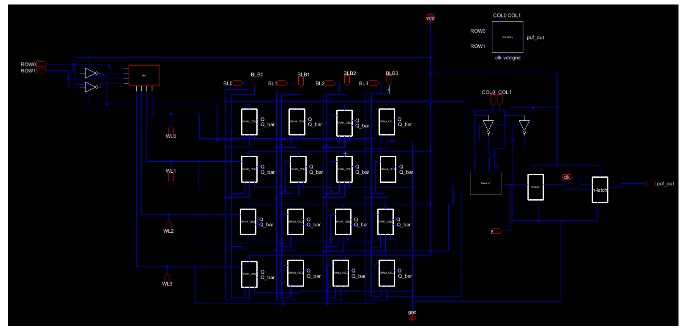

Fig. 3. Schematic implementation of the proposed 4×4 SRAM PUF array.

F. 4×4 PUF Array

To generate multiple PUF response bits simultaneously, the proposed architecture extends the single SRAM PUF cell into a 4×4 array configuration. The array consists of sixteen SRAM PUF cells organized in rows and columns. This arrangement enables the generation of multiple independent response bits while maintaining a compact and scalable architecture.

Figure 3 shows the hierarchical schematic implementation of the proposed 4×4 SRAM PUF array, where sixteen SRAM PUF cells are organized in a row-column architecture to generate multiple response bits simultaneously.

Figure 4 presents the layout implementation of the proposed 4×4 SRAM PUF array designed using the Electric VLSI Design System. The layout arranges the sixteen SRAM PUF cells in a regular matrix while providing organized routing for the word lines, bit lines, and power connections. This hierarchical implementation improves the scalability of the proposed architecture and facilitates its integration into larger PUF designs.

G. Unstable-Bit Detection Circuit

Since some SRAM cells may produce inconsistent responses under repeated evaluations, an unstable-bit detection circuit is included in the proposed architecture. This circuit identifies cells that exhibit unstable behavior and distinguishes

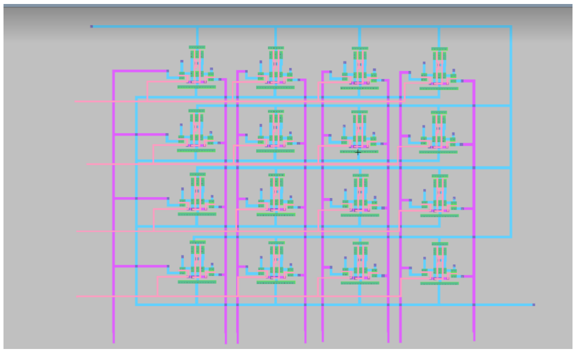

Fig. 4. Layout implementation of the proposed 4×4 SRAM PUF array.

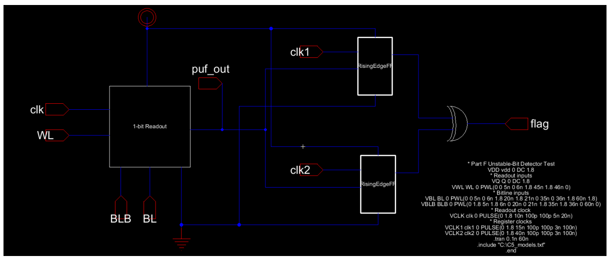

Fig. 5. Schematic implementation of the unstable-bit detection circuit.

them from reliable cells. Detecting unstable bits improves the overall reliability of the generated hardware fingerprint and supports the evaluation of the proposed SRAM PUF architecture under different operating conditions.

Figure 5 illustrates the schematic implementation of the unstable-bit detection circuit. The detector monitors the generated PUF response over repeated evaluations and produces a flag indicating whether the response is stable or unstable. This information is later used by the masking and stabilization stage to improve the reliability of the final PUF output.

H. Masking and Stabilization

The masking and stabilization stage represents the final processing stage of the proposed SRAM PUF architecture. Although the Schmitt trigger improves the stability of the readout process, some response bits may still remain unstable due to process variations and environmental noise. Therefore, a masking stage is introduced to identify and discard these unreliable bits. The stage combines the outputs of the 4×4 PUF array with the flag generated by the unstable-bit detection circuit. Using this flag, unstable response bits are masked, while only the stable response bits are preserved to construct the final PUF output. This process improves the repeatability and overall reliability of the generated hardware fingerprint without modifying the original PUF generation mechanism.

Figure 6 illustrates the masking and stabilization stage of the proposed design. The 4×4 PUF array generates the initial PUF response, while the unstable-bit detector analyzes the response and produces a flag indicating unreliable bits. The masking stage then removes the flagged unstable bits and preserves only the reliable response bits, resulting in a more stable and repeatable PUF output.

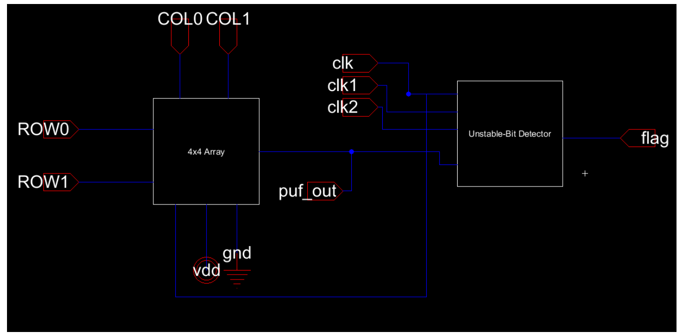

Fig. 6. Schematic implementation of the masking and stabilization stage.

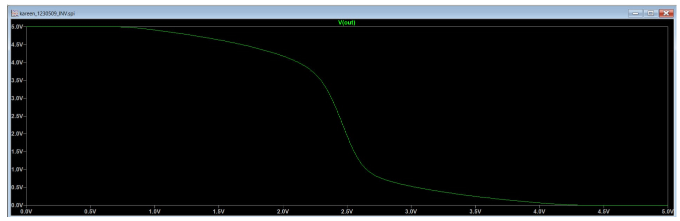

Fig. 7. Simulated voltage transfer characteristic (VTC) of the designed CMOS inverter.

IV. Simulation Setup

All circuit blocks were designed using the Electric VLSI Design System. Both schematic and layout implementations were completed for each circuit, while functional verification was performed through SPICE transient simulations using the C5 CMOS transistor model library. The simulations were carried out to verify the functionality of each block before integrating them into the complete PUF architecture.

V. Results and Discussion

A. CMOS Inverter Results

Figure 7 presents the simulated voltage transfer characteristic (VTC) of the designed CMOS inverter. The simulation demonstrates the expected CMOS inverter behavior, where the output remains at a logic-high level for low input voltages and switches to a logic-low level as the input voltage increases. The sharp transition between the two logic levels confirms the correct switching operation of the PMOS and NMOS transistors.

The switching threshold voltage (V_M) obtained from the SPICE simulation is approximately **2.44 V**, which is close to the midpoint of the 5 V supply voltage. This indicates that the inverter exhibits nearly symmetrical switching characteristics and confirms the proper sizing and complementary operation of the PMOS and NMOS transistors.

Therefore, the simulation verifies that the designed CMOS inverter is functioning correctly and is suitable for use as the fundamental building block of the proposed SRAM PUF architecture.

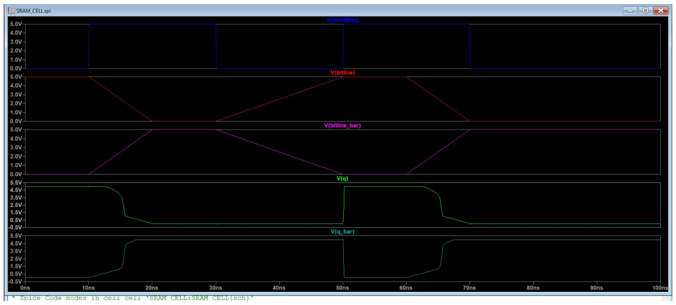

Fig. 8. Transient simulation of the conventional 6T SRAM cell showing the word line (WL), bit lines (BL and BLB), and the internal storage nodes (Q and $\overline{Q}$).

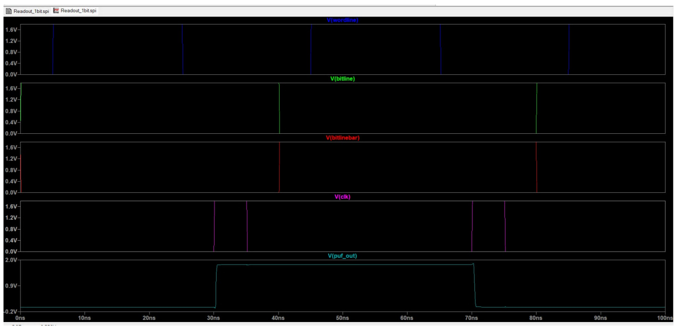

Fig. 9. Transient simulation of the Schmitt trigger-assisted SRAM PUF readout circuit.

B. Conventional 6T SRAM PUF Results

Figure 8 presents the transient simulation results of the conventional 6T SRAM cell. The applied word line (WL) controls the access transistors, allowing data to be written through the complementary bit lines (BL and BLB). The internal storage nodes Q and $\overline{Q}$ switch to complementary logic levels during the write operation and preserve the stored state after the word line is deactivated. This behavior confirms the correct operation of the cross-coupled inverters and demonstrates that the SRAM cell successfully performs both write and hold operations. Therefore, the designed conventional 6T SRAM cell operates correctly and provides the basic memory element required for the proposed SRAM PUF architecture.

C. One-Bit PUF Readout Circuit Results

Figure 9 shows the transient simulation of the Schmitt trigger-assisted SRAM PUF readout circuit. Similar to the conventional SRAM cell, the memory operation is controlled by the word line (WL) and complementary bit lines (BL and BLB). However, unlike the conventional design, the stored value is processed through the Schmitt trigger readout circuit to generate the final *PUF_OUT* signal. The simulation shows that the output transitions between logic levels are clean and well-defined, producing a stable digital output during the read operation. Compared with the conventional SRAM cell shown in Fig. 8, the proposed design provides a clearer readout signal and improves the robustness of the sensing process, making it more suitable for reliable PUF response generation.

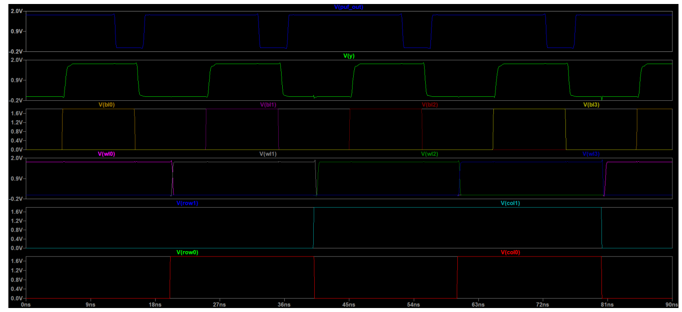

Fig. 10. Transient simulation of the proposed 4×4 SRAM PUF array.

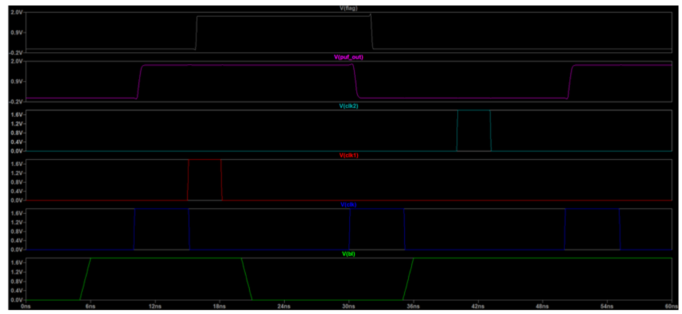

Fig. 11. Transient simulation of the unstable-bit detection circuit.

D. 4×4 PUF Array Results

Figure 10 presents the transient simulation results of the proposed 4×4 SRAM PUF array. The simulation demonstrates the coordinated operation of the row and column selection signals together with the corresponding word lines and bit lines. During each access cycle, the selected SRAM PUF cell generates the corresponding response, which appears at the PUF output. The waveforms show that only the addressed cell is accessed at a given time, while the remaining cells remain inactive. This confirms the correct operation of the array organization and verifies that the proposed architecture successfully extends the single-bit PUF cell into a scalable multi-bit PUF implementation capable of generating multiple independent response bits.

E. Unstable-Bit Detection Results

Figure 11 presents the transient simulation results of the unstable-bit detection circuit. The simulation shows the generated PUF response together with the control clock signals and the corresponding flag output. During the evaluation process, the detector monitors the PUF response over the applied clock cycles and generates a flag signal according to the detected response stability. The obtained waveforms confirm the correct operation of the unstable-bit detection circuit, demonstrating its ability to identify unreliable responses and provide the information required for the subsequent masking and stabilization stage.

F. Performance Evaluation and Comparison

The implementation results demonstrate that the proposed SRAM PUF architecture achieves improved reliability and stability by incorporating the Schmitt trigger-assisted readout circuit. Although the additional circuitry increases the silicon area, it significantly enhances the robustness of the generated PUF responses by reducing the effect of noise and small voltage fluctuations. Furthermore, the proposed design maintains low power consumption while providing a more reliable hardware fingerprint, making the additional hardware overhead an acceptable trade-off for secure hardware authentication applications.

TABLE I
IMPLEMENTATION RESULTS OF THE PROPOSED SRAM PUF ARCHITECTURE.

Parameter	**Value**
SRAM Cell Area	7,812
Schmitt-Assisted Readout Area	114,758
4×4 SRAM PUF Array Area	523,349
SRAM Average Power	0.502 mW
Schmitt-Assisted Readout Average Power	1.683 μW
4×4 Array Average Power	17.56 μW

As shown in Table I, the proposed architecture introduces additional hardware to improve the reliability of the generated PUF responses. The area of the Schmitt-assisted readout circuit increases from 7,812 for the conventional SRAM cell to 114,758 due to the integration of the Schmitt trigger and the supporting readout circuitry. Furthermore, the complete 4×4 SRAM PUF array occupies a total area of 523,349, demonstrating the scalability of the proposed design.

In terms of power consumption, the conventional SRAM cell consumes an average power of 0.502 mW, whereas the Schmitt-assisted readout circuit consumes only 1.683 μW. The complete 4×4 array consumes 17.56 μW while integrating sixteen SRAM cells together with the additional control circuitry. These results indicate that the proposed architecture successfully improves the stability and noise immunity of the generated PUF responses while maintaining low power consumption, making the increase in implementation area an acceptable trade-off for reliable hardware authentication.

VI. CONCLUSION

This paper presented the design and implementation of a Schmitt trigger-assisted SRAM Physical Unclonable Function (PUF) for hardware authentication applications. Starting from a conventional 6T SRAM cell, the proposed architecture integrates a Schmitt trigger-assisted readout circuit, a 4×4 SRAM PUF array, an unstable-bit detection block, and a masking stage to improve the reliability of the generated PUF responses. The complete design was implemented and verified using Electric VLSI and LTspice simulations.

The implementation results show that the proposed architecture provides improved response stability and stronger immunity to noise while maintaining low power consumption. Although the additional circuitry increases the implementation area, this overhead represents a reasonable trade-off for achieving more reliable hardware fingerprints. Overall, the proposed SRAM PUF architecture demonstrates the feasibility of enhancing conventional SRAM-based PUFs through simple and effective stabilization techniques, making it a promising solution for secure hardware authentication.

REFERENCES

[1] Y. He, D. Li, Z. Yu, and K. Yang, "ASCH-PUF: A Zero Bit Error Rate CMOS Physically Unclonable Function with Dual-Mode Low-Cost Stabilization," *IEEE Journal of Solid-State Circuits*, 2023.